

CivicPulse

AI-Powered Spatial Planning, Multi-Modal Accessibility Analytics & Urban Infrastructure Resilience Platform

A complete, production-grade guide covering system architecture, mathematical formulas, What-If simulation mechanics, live demo scripts, and judges' technical defense.

PROJECT VERSION

v0.5.0 (Stage 12 Complete)

AUTOMATED TESTS

157 / 157 Passing (100%)

FRONTEND BUNDLE

React 19 + Leaflet + Vite

CORE BACKEND

FastAPI + Python 3.12

GEOSPATIAL ENGINE

Shapely + PostGIS / SQLite

MAP TILES & LAYER

OpenStreetMap (Zero API Cost)

Designed & Built for Municipal Authorities, Urban Planners, and Engaged Communities.

1. Executive Summary & Problem Statement

The Core Problem: In rapidly growing urban centers, essential public services—clinics, schools, clean water points, transit stops, and food markets—are distributed inequitably. Vulnerable neighbourhoods become **service deserts** where residents travel over 45 minutes for basic healthcare. Meanwhile, municipal authorities allocate multi-million dollar infrastructure budgets using subjective guesswork, political intuition, or static census spreadsheets, while citizen failure reports remain trapped in disconnected silos.

CivicPulse is an open-standard, mathematically grounded urban spatial intelligence platform that bridges top-down municipal planning with bottom-up citizen reality. Rather than relying on opaque black-box AI models that hallucinate allocations, CivicPulse provides:

1. Deterministic Multi-Criteria Analytics

Synthesizes travel time, facility throughput load, transport connectivity, and demographic vulnerability into reproducible 0–100 accessibility and service desert scores with transparent mathematical formulas.

2. Explainable Candidate Allocation

Identifies optimal intervention locations using geometric centroid, population density, and gap-perimeter strategies. Every recommendation includes human-auditable justification factors.

3. In-Memory What-If Simulation Lab

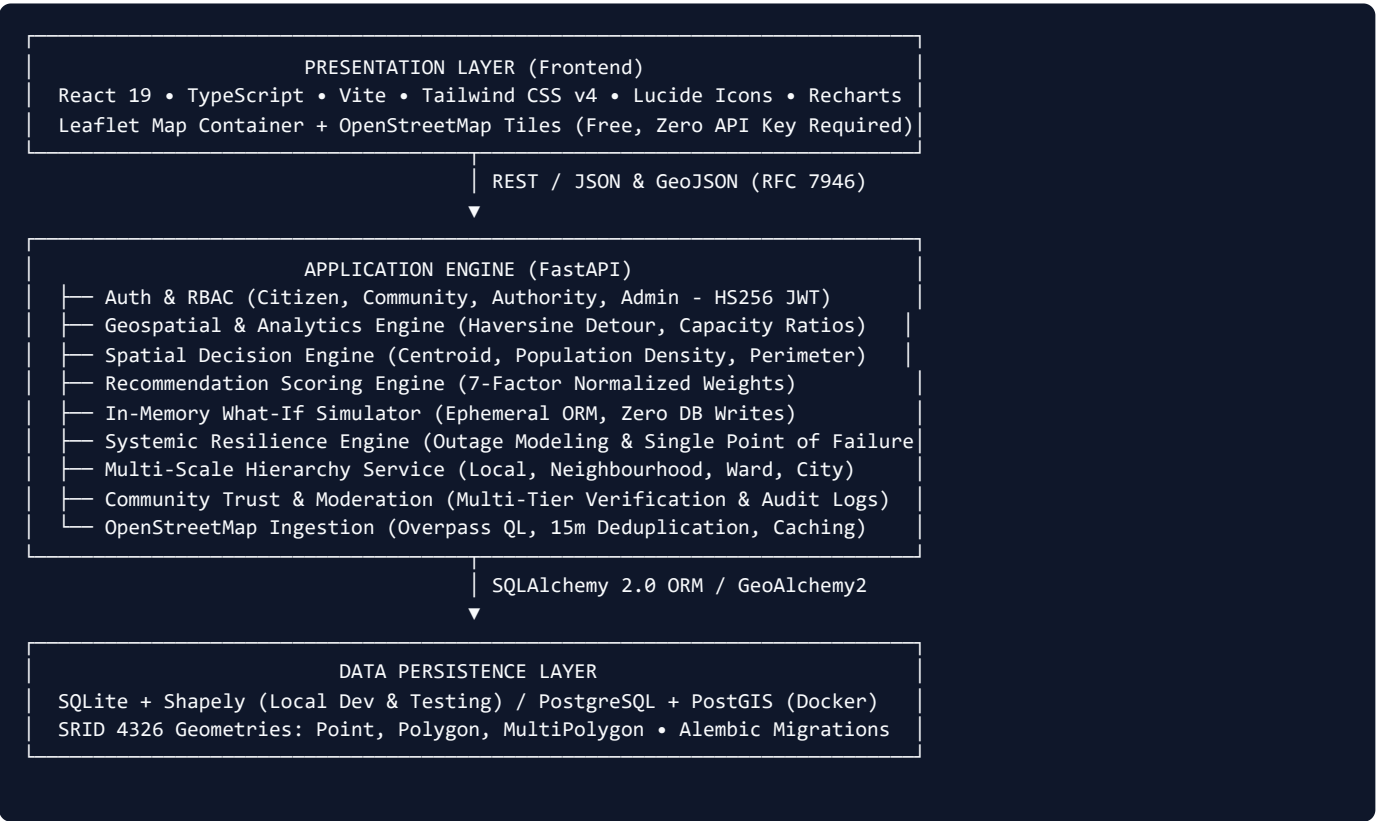
Enables planners to simulate building or upgrading facilities before committing taxpayer capital. Calculates accessibility gains and population relieved in-memory with **zero database writes**.

4. Verified Civic Trust Hierarchy

Measures the *Reality Gap* by tracking real-world infrastructure failures submitted by citizens through an immutable, role-guarded verification pipeline from community review to official confirmation.

2. System Architecture & Tech Stack

CivicPulse is built with a strictly decoupled, production-ready micro-architecture adhering to clean architecture patterns:



COMPONENT	TECHNOLOGY	ARCHITECTURAL RATIONALE
Backend Framework	Python 3.12, FastAPI, Pydantic v2	High-concurrency async ASGI server, automated OpenAPI 3.1 documentation, strict data contract validation.
Spatial ORM & Geometry	SQLAlchemy 2.0, GeoAlchemy2, Shapely	Standardized WKT and GeoJSON geometry handling (SRID 4326), robust polygon containment and centroid calculations.
Database Engine	SQLite (local) / PostgreSQL + PostGIS (prod)	Zero-setup lightweight deterministic local execution; transparent production scaling via PostGIS spatial indexes.
Frontend Web App	React 19, TypeScript, Vite, Tailwind v4	Sub-second build times (1.5s), strictly typed data models, reactive state management, modern civic-tech UI design.
Interactive Map	React-Leaflet, Leaflet 1.9, OpenStreetMap	100% free, key-free tile rendering adhering to OSM usage policies, custom SVG divlcon markers, GeoJSON choropleth layers.
Security & Auth	OAuth2 Password Bearer, JWT, bcrypt (12 rounds)	Stateless authentication with server-side Role-Based Access Control (RBAC) enforced on sensitive mutation endpoints.

3. Core Geospatial & Analytical Engines

Unlike simplistic GIS applications that merely draw 500-meter radius circles around facilities, CivicPulse computes a multi-dimensional accessibility score grounded in urban reality.

A. Composite Accessibility Score Formula

$$Accessibility\ Score = 0.30 \times S_{travel} + 0.20 \times S_{avail} + 0.20 \times S_{capacity} + 0.15 \times S_{conn} + 0.15 \times S_{equity}$$

$$Gap\ Score = 100.0 - Accessibility\ Score$$

FACTOR (WEIGHT)	DESCRIPTION & IMPLEMENTATION	MATHEMATICAL MECHANICS
Travel Time (30%)	Measures time required to reach the nearest facility under urban conditions.	Haversine great-circle distance × 1.30 urban street grid detour coefficient. Walking speed: 4.5 km/h; Transit speed: 20 km/h. Decays from 100 (instant access) to 0 (>45 mins).
Service Availability (20%)	Operational condition of the target infrastructure facility.	Categorical weighting: operational = 100, limited = 60, degraded = 50, temporarily_unavailable = 20, closed = 0.
Capacity Pressure (20%)	Demand vs available throughput throughput ratio (D/C).	Calculates demand from neighbourhood population vs facility throughput. Ratios > 1.5 indicate critical overcrowding; ratios < 0.7 indicate comfortable capacity.
Transport Connectivity (15%)	Proximity to multimodal transit nodes (metro, bus terminals).	Evaluates whether the facility is reachable via public transit, scoring 100 if within 400m of a transit stop.
Demographic Equity (15%)	Prioritization of socio-economically vulnerable areas.	Derived from fine-grained population cells factoring elderly ratio, child dependency, and vulnerability indexes.

B. Service Desert Classification

Every administrative area is classified deterministically based on its composite accessibility score:

SCORE RANGE	CLASSIFICATION	MAP COLOR	PLANNING IMPLICATIONS
80 – 100	WELL SERVED	Emerald Green	Optimal coverage; low travel times; adequate facility capacity. Maintenance only.
60 – 79	ADEQUATE	Yellow	Acceptable baseline access, but vulnerable to seasonal demand spikes or facility degradation.
40 – 59	AT RISK / UNDERSERVED	Amber / Orange	Emerging service deficit; excessive travel times (>25 mins) or capacity pressure > 100%.
20 – 39	UNDERSERVED	Deep Orange	Acute infrastructure deficit; high equity vulnerability; priority candidate for municipal capital.
0 – 19	CRITICAL DESERT	Crimson Red	Severe civic deprivation; zero operational facilities within standard catchment radius.

C. Pluggable Routing Engine

CivicPulse decouples travel estimation using a pluggable `RoutingProvider` architecture:

- **Deterministic Urban Detour Model (Default / Offline):** Scales great-circle distance by a calibrated \$1.30 times Manhattan grid coefficient. Executes in sub-milliseconds without incurring external API costs.
- **OSRM (Open Source Routing Machine):** When `USE_OSRM=True`, queries road network graph servers. If OSRM times out or is unreachable, the system automatically falls back to the deterministic model with an audit notice.

4. Spatial Decision Engine & Recommendation System

When a locality is flagged as a service desert, the planner needs to know: *"Where exactly should we intervene, and why?"* CivicPulse answers this algorithmically.

A. Algorithmic Candidate Location Allocation

Rather than requiring planners to manually click random map coordinates, the `CandidateGenerator` generates candidate sites using three complementary geometric strategies:

1. Geometric Centroid

Computes the interior mathematical centroid of the underserved neighbourhood polygon using Shapely. Represents the geometric center of gravity.

2. Population Density Node

Locates the centroid of the population cell with the highest resident density. Maximizes walking-catchment population within 15 minutes.

3. Gap Perimeter Node

Identifies the interior coordinate furthest from all existing facilities in the category, eliminating geographic blind spots.

Validation & Deduplication

All candidates are verified for strict geometric containment inside the boundary (`polygon.contains(pt)`) and spatially deduplicated within an 11-meter tolerance.

B. 7-Factor Recommendation Scoring Formula

Every valid candidate location is evaluated across seven normalized criteria (0 to 100), producing an auditable priority ranking:

$$Score = 0.30 \times Gap + 0.25 \times Pop + 0.15 \times Travel + 0.10 \times Capacity + 0.10 \times Equity + 0.05 \times Connectivity + 0.05 \times Confidence$$

FACTOR	WEIGHT	ANALYTICAL RATIONALE
Gap Severity	30%	Prioritizes interventions in localities with the highest baseline deficit.
Population Affected	25%	Scales civic ROI by the absolute number of residents benefiting from the new site.
Travel-Time Need	15%	Targets areas with excessive existing commute times to essential amenities.
Capacity Pressure	10%	Relieves overburdened neighboring clinics/schools operating above 100% capacity.
Equity Need	10%	Directs resources to historically marginalized or vulnerable demographic zones.
Transport Connectivity	5%	Rewards sites located along transit corridors to ensure regional access.
Data Confidence	5%	Weights decisions by the statistical reliability and verification level of inputs.

C. Natural Language Explainability (XAI)

To ensure accountability in public hearings, every scored candidate generates human-readable justification strings explaining the mathematical drivers:

"Top Ranked Candidate (cand-healthcare-9-centroid): Ranked #1 (Score: 80.6) because of severe healthcare accessibility gap (86.9%) in Highlands Valley, large affected population (22,000 residents), long estimated travel time with no reachable facility in catchment, and strong transit connectivity supporting regional catchment."

5. In-Memory What-If Simulation Lab & Resilience Analysis

The What-If Simulation Lab is CivicPulse's core planning innovation. It allows urban planners and municipal budget committees to test prospective capital expenditures before spending public funds.

Strict Safety & Zero Database Writes Guarantee: The simulation engine runs **100% in-memory**. Proposed facilities are instantiated as ephemeral ORM instances. Zero `session.add()` or `session.commit()` operations occur. Planners can run dozens of simulations without risking data corruption or schema mutations.

A. Before vs After Delta Evaluation

When a planner triggers a simulation for a candidate intervention, the engine executes a comparative dual evaluation:

- Baseline Calculation:** Evaluates current locality accessibility, citywide coverage %, underserved population count, and average travel time.
- Synthetic Blending:** Dynamically injects the proposed facility into catchment routing queries in-memory.
- Delta Computation:** Measures exact marginal gains:
 - $\Delta \text{Accessibility Score} = \text{Score}_{\text{after}} - \text{Score}_{\text{before}}$
 - $\Delta \text{Citywide Coverage \%} = \text{Coverage}_{\text{after}} - \text{Coverage}_{\text{before}}$
 - $\text{Underserved Population Relieved} = \text{Residents transitioning out of Desert status}$
 - $\text{Travel Time Saved} = \text{Baseline Minutes} - \text{Simulated Minutes}$

Metric	Baseline (Highlands Valley)	Post-Simulation State	Measured Civic Impact
Accessibility Score	13.1 / 100 (Critical Desert)	78.8 / 100 (Well Served)	+65.7 Points
Citywide Healthcare Coverage	68.2%	92.1%	+23.9% Expansion
Underserved Residents	22,000 residents	0 residents	22,000 Residents Relieved
Average Travel Time	60.0 minutes (transit)	2.1 minutes (walk)	57.9 Minutes Saved per Trip

B. Systemic Resilience & Single Point of Failure (SPOF) Analysis

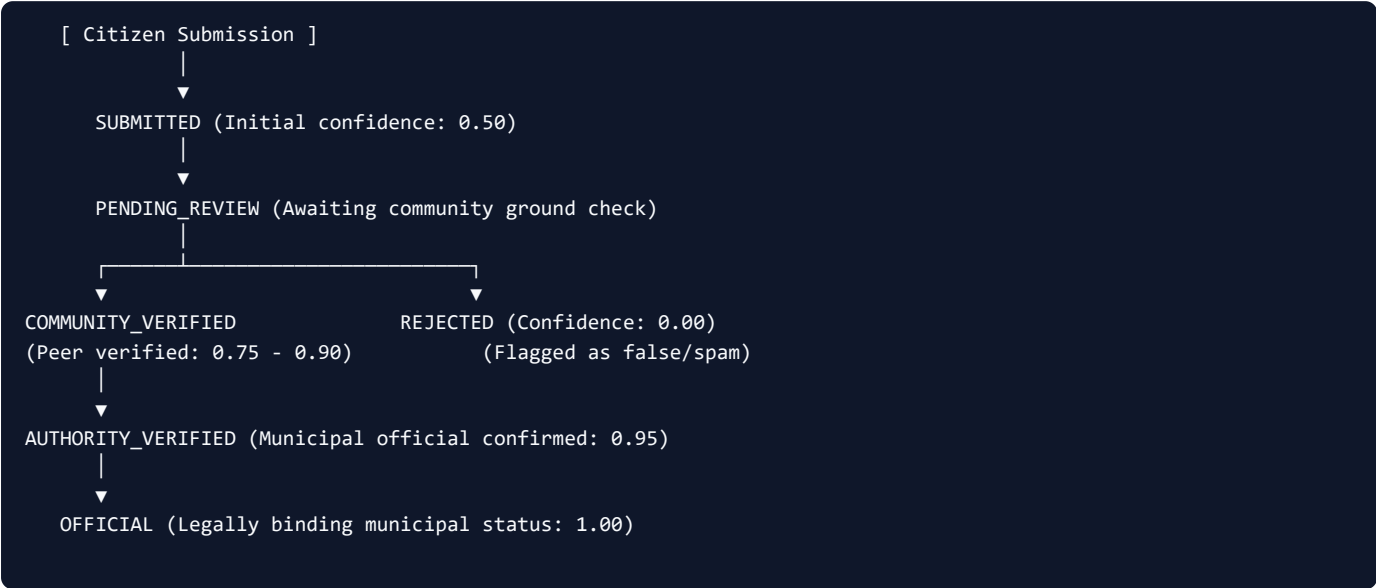
In addition to building new infrastructure, CivicPulse models disaster scenarios:

- Facility Failure Modeling:** Simulates an existing hospital or water treatment plant going offline due to flood, grid failure, or strike.
- Cascading Impact Evaluation:** Quantifies the resulting coverage drop and flags whether the facility is a **Single Point of Failure (SPOF)** whose collapse plunges downstream neighbourhoods into critical desert status.

6. Community Ground Truth & Civic Trust Hierarchy

Official municipal databases frequently present an idealized reality: a water borehole or primary clinic may be cataloged as "Operational," but on the ground it is padlocked, flooded, or lacks power. CivicPulse bridges this disparity with the **Reality Gap Index** and a multi-tiered verification pipeline.

A. Multi-Tier Verification Lifecycle



B. Server-Side Role-Based Access Control (RBAC)

Civic trust cannot rely on client-side UI checks. CivicPulse enforces strict server-side RBAC guards:

ROLE	SYSTEM PERMISSIONS	VERIFICATION AUTHORITY
citizen	Public read (data:read), submit report (report:create).	Cannot approve or verify reports (Attempts return HTTP 403 Forbidden).
community	Citizen permissions + peer community check (report:verify_community).	Can conduct peer verification up to COMMUNITY_VERIFIED status.
authority	Municipal operations (authority:operate) + official audit (report:verify_official).	Can promote reports to OFFICIAL status or mark as REJECTED.
admin	Platform control (admin:manage), role assignment, and audit oversight.	Full administrative moderation authority across all system states.

C. Immutable Audit Logging

Every status modification is captured in an append-only AuditLog table storing the actor ID, entity ID, prior status, new status, audit reason, and UTC timestamp. This guarantees full transparency for legal compliance and municipal review.

7. Dual Operational Modes & OpenStreetMap Ingestion

DEMO MODE (Default & Offline)

- Self-contained deterministic dataset of "Metro City" (Bangalore coordinates).
- 10 administrative areas, 14 verified facilities across 5 categories, 50 community reports.
- Operates 100% offline without API keys or external network dependencies.
- Guarantees zero presentation flakiness during hackathon demos.

REAL DATA MODE (Live Global GIS)

- Queries OpenStreetMap via live Overpass QL API for any global municipality.
- Extracts 5 core service categories: Healthcare, Education, Transport, Water, Markets.
- Anti-Null Island coordinate sanitizer (rejects 0,0 or out-of-bound coordinates).
- 15-meter spatial deduplication buffer against existing database records.

Strict Population Data Integrity Principle: When ingesting real-world geographic data, CivicPulse **never invents, hallucinates, or synthesizes census population figures**. If documented public census data is provided, it is stored as `population_status="documented"`. If census records are unavailable, the count is recorded as `None` with `population_status="unavailable"`, and downstream capacity formulas adapt gracefully with explicit data quality notices.

8. Step-by-Step 3-Minute Live Hackathon Demo Walkthrough

Use this exact script and click sequence during the presentation to deliver a punchy, high-impact demo:

- 1 Problem Hook (0:00 – 0:30)**
"Judges, urban inequality isn't just about income—it's about access. In every city, vulnerable neighbourhoods become service deserts where reaching a clinic takes an hour or more. Planners make multi-million dollar capital decisions using static spreadsheets, while citizen reports remain ignored. We built CivicPulse to bridge top-down municipal planning with bottom-up citizen reality."
- 2 Interactive Map & Locality Service Gap (0:30 – 1:00)**
Action: Open <http://127.0.0.1:5173/map>. Point out OpenStreetMap tiles and custom color-coded category markers. Click on **Highlands Valley** polygon.
"On the CivicPulse map, we see the city's administrative boundaries and cataloged facilities across healthcare, education, transport, water, and food markets. Clicking Highlands Valley reveals a critical healthcare gap: Accessibility Score is 13.1 / 100, Gap is 86.9, and the nearest clinic is 7.8 km away, representing an estimated 60-minute travel time."
- 3 Planner Command Center & Recommendations (1:00 – 1:40)**
Action: Log in as Municipal Authority (authority@example.com / Authority123!). Switch to Recommendations Layer or open Planner view.
"Logging in as Municipal Authority unlocks planning intelligence. The priority leaderboard instantly ranks Highlands Valley at #1. Instead of guessing where to build, we click Generate Recommendations. Our spatial engine scores candidate locations across 7 normalized criteria. Candidate cand-healthcare-9-centroid ranks #1 with clear explainable justifications."
- 4 What-If Simulation Lab & Measured Impact (1:40 – 2:20)**
Action: Run the simulation for the top candidate.
"Now for our core innovation: the What-If Simulation Lab. Before spending public funds, the planner simulates building a community health center at this exact site. In milliseconds, our in-memory engine re-evaluates the city: Locality accessibility jumps from 13.1 to 78.8 (+65.7 points)! 22,000 residents are relieved from a healthcare desert! Average travel time drops from 60 minutes to 2.1 minutes! And zero database writes occurred—planners can test dozens of scenarios completely risk-free."
- 5 Community Ground Truth & OpenStreetMap Toggle (2:20 – 3:00)**
Action: Show Community Reports layer and toggle Query OSM.
"Finally, CivicPulse integrates citizen ground reality. Citizens report infrastructure breakdowns; community members verify them; municipal officials approve them as Official with an immutable audit log. And with a single click on 'Query OSM', we can ingest live OpenStreetMap amenities globally with automated deduplication. CivicPulse makes urban planning transparent, equitable, and data-driven. Thank you!"

9. Judges' Technical Q&A & Defense Cheat Sheet

JUDGE QUESTION	MODEL TECHNICAL DEFENSE ANSWER
Q1: Why use deterministic scoring rather than an LLM or deep learning?	Municipal capital allocations involve statutory audits, legal liabilities, and public tax dollars. A city council cannot defend an infrastructure decision with an unexplainable neural network that hallucinates or produces non-reproducible outputs. CivicPulse uses mathematically verifiable formulas with exact factor weights ($\$0.30 \times \text{Gap} + 0.25 \times \text{Pop} + \dots$). Every score provides auditable mathematical proof.
Q2: How do you estimate travel times without huge Google Maps API bills?	We built a pluggable <code>RoutingProvider</code> architecture. By default, it uses a deterministic urban detour model scaling Haversine distances by an empirical $\$1.30 \times$ street grid factor with walking and transit speed profiles. It executes in sub-milliseconds, runs 100% offline, and costs $\$0.00$. For live network graphs, it integrates with open-source OSRM with automatic fallback.
Q3: How do you prevent citizens from spamming fake reports?	We enforce a 4-tier verification pipeline: citizen reports enter as <code>PENDING_REVIEW</code> (confidence 0.50). Peer community review raises trust to 0.75–0.90. Only verified municipal authorities can approve <code>OFFICIAL</code> status (confidence 1.00) or reject false reports. Server-side RBAC returns HTTP 403 if unauthorized users attempt official approvals, and every action is recorded in an immutable <code>audit_logs</code> table.
Q4: In Real Data Mode, what happens if census population data is missing? Do you synthesize it?	Never. We adhere to a strict data integrity principle: never present synthetic values as real measurements. If population data is missing, we record <code>population_status="unavailable"</code> and report capacity pressure as neutral with an explicit data quality flag. We never fabricate numbers.
Q5: How do simulations guarantee database integrity?	Simulations run entirely in-memory . Proposed facilities are instantiated as ephemeral ORM objects in Python memory. The analytics engine blends them into catchment calculations on the fly. Zero <code>session.add()</code> or <code>session.commit()</code> calls execute, ensuring 100% database schema and data safety.
Q6: How do you handle OpenStreetMap licensing and usage limits?	All Overpass data adheres to the Open Database License (ODbL 1.0). Every record stores explicit attribution, query timestamp, and raw OSM tags. Ingestion queries are hashed via SHA256 with a 24-hour TTL cache, and we enforce a 1.0s query cooldown to respect public server usage policies.
Q7: How does the backend tech stack scale to millions of residents?	FastAPI runs asynchronous non-blocking request loops. On the database layer, transitioning from SQLite to PostgreSQL + PostGIS unlocks spatial GiST indexing (R-Tree indexes), enabling spatial queries across millions of coordinates in milliseconds.
Q8: What are current limitations and future roadmap?	Currently, simulations evaluate single facilities or defined multi-facility lists; combinatorial optimization across hundreds of simultaneous budget permutations is computationally intensive and planned for Stage 12+. Real-time updates currently use REST polling; WebSockets are slated for live pin broadcasting.

10. Verification, Credentials & Presentation Checklist

System Verification Metrics

- **Automated Backend Tests:** 157 / 157 Passing (100%)
- **Frontend Production Build:** tsc -b && vite build passes cleanly (1.51s build time)
- **Code Linting:** oxlint passes with 0 errors across 31 files and 116 rules
- **Backend Health Probes:** GET /health returns healthy with SQLite connected

Demo Account Credentials

- **Citizen:** citizen@example.com / Citizen123!
- **Community:** community@example.com / Community123!
- **Municipal Authority:** authority@example.com / Authority123!
- **Admin:** admin@example.com / Admin123!

Quick Launch Commands for Demo Day:

backend\.venv\Scripts\python.exe backend un.py | cd frontend && npm.cmd run dev

Backend: **http://127.0.0.1:8000** | Frontend: **http://127.0.0.1:5173**